AHAN CHAKRABORTY

Kolkata, West Bengal, India
Curriculum Vitae
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Education

- **B.Stat** — 2024 – Present
  Indian Statistical Institute, Kolkata — *Kolkata*
  - Percentage: 97.1 (till present)
- **Higher Secondary Education** — 2022 – 2024
  Ramakrishna Mission Boys' Home High School, Rahara — *Kolkata*
  - Percentage: 97.6
  - Secured a rank of 9 in the state (out of ∼7,60,000 students)
- **Secondary Education** — 2012 – 2022
  Ramakrishna Mission Boys' Home High School, Rahara — *Kolkata*
  - Percentage: 97.2
  - Secured a rank of 13 in the state (out of ∼11,00,000 students)

Experience / Projects

- **Benign Overfitting without Linearity, Hyperbolic Neural Networks** — June 2026 – Present
  ISI Kolkata — *(online)*
  - **Supervisor:** Prof. Saptarshi Chakraborty, University of Michigan.
  - **Brief Description:** Studying the papers "Benign Overfitting without Linearity: Neural Network Classifiers Trained by Gradient Descent for Noisy Linear Data" by Bartlett, Chatterji, Frei and "Hyperbolic Neural Network" by Bécigneul, Ganea, Hofmann.
- **Random Cluster Model on Regular Graphs** — May 2026 – June 2026
  SN Bhatt Memorial Excellence Fellowship Programme, ICTS Bangalore, TIFR
  - **Supervisor:** Prof. Anirban Basak, Prof. Ridhdhipratim Basu, ICTS Bangalore.
  - **Brief Description:** Explored various probabilistic aspects regarding Statistical Mechanics such as models like Ising, Potts, Random Cluster, Percolation; methods like Belief Propagation. Concluded with reading and presenting the paper "Random Cluster Model on Regular Graphs" by Ferenc Bencs, Márton Borbényi & Péter Csikvári.
- **An Introductory Reading Project in Causal Inference** — May 2026 – Present
  ISI Kolkata — *(online)*
  - **Supervisor:** Prof. Ambarish Chatterjee, ISI Kolkata.
  - **Brief Description:** Introductory reading project on Causal Inference, exploring basic concepts from "Causal Inference: A Statistical Learning Approach" by Stefan Wager.
- **Clustering Methods and Performance Comparison in NCI60 Dataset** — April 2026
  ISI Kolkata — *(class project)*
  - **Supervisor:** Prof. Shyamal Krishna De, ISI Kolkata.
  - **Brief Description:** Implementing various clustering algorithms like standard K-means, hierarchical, and newer algorithms like Convex, Sparse Convex, Sparse Hierarchical. Presented a paper on K-means with MAD distance by S.K.De, A.K.Ghosh, B.Paul, part of a class project.

- **Random Graphs and Limit Theory for Sparse Random Graphs** Dec 2025 – April 2026
  ISI Kolkata
  - **Supervisor:** Prof. Soumendu Sundar Mukherjee, ISI Kolkata.
  - **Brief Description:** Explored probabilistic concepts, including branching processes, coupling, the probabilistic method, phase transitions, connectivity, sub-critical and super-critical regimes, subgraph containment thresholds in Erdős-Rényi random graphs, and Benjamini-Schramm Convergence from standard texts (Alon-Spencer, Remco Hofstad, Karoński-Frieze, Roch, Rick Durrett).
- **A Reading Project on Convex Clustering** Nov 2025 – Feb 2026
  ISI Kolkata *(online)*
  - **Supervisor:** Prof. Saptarshi Chakraborty, University of Michigan.
  - **Brief Description:** Conducted a comprehensive reading project on SVM, traditional clustering techniques, and Convex clustering methods.
- **Analysis of Habitability of Exoplanets using Learning Tools** Oct 2025 – Present
  ISI Kolkata
  - **Supervisor:** Prof. Smarajit Bose, Prof. Asis Chatterjee, Prof. Amita Pal, ISI Kolkata.
  - **Brief Description:** Extracted key habitability features and predicted the habitability of exoplanets utilizing machine learning tools.
- **A Reading Project in Statistical Learning** May 2025 – Aug 2025
  ISI Kolkata
  - **Supervisor:** Prof. Smarajit Bose., ISI Kolkata.
  - **Brief Description:** Explored statistical learning tools, linear programming, and convex optimization through foundational texts (e.g., ESL, ISLR, LTFP Statistical Pattern Recognition by Fukunaga, Convex Optimization by Boyd, and Linear Programming by Matousek). Additionally, reviewed research papers on Generalized QDA and hub-awareness in KNN classifiers.
- **2025 Summer School in Statistics for Astronomers** June 2025
  Center for Astrostatistics at The Pennsylvania State University. *(online)*

Internships

SN Bhatt Memorial Excellence Fellowship, ICTS Bangalore, TIFR Summer 2026
Mentor: Prof. Anirban Basak, Prof. Ridhdhipratim Basu.

Academic Accolades

- Received prize money for excellent academic performance in each semester till present.
- Secured Rank 5 All India in Madhava Mathematics Competition, 2025.
- University topper in Simon Marais Mathematics Competition, 2024; Rank 13 in East Division, Rank 4 in India (in pair with Arnab Sanyal), received Jane Street cash prize.
- Secured Rank 12 in B.Stat Entrance 2024, ISI Kolkata (out of ∼15,000 candidates).
- Received SHRIRAM Scholarship in CMI Entrance, 2024 (rejected, ∼50 students were given the scholarship across India).
- Secured All India Rank 1227 in JEE Advanced (IIT), 2024 (out of ∼1.8 lakh candidates).
- Secured Rank 15 in WBJEE, 2024 (out of ∼1 lakh candidates).
- Qualified Regional Math Olympiad, 2023.
- Received *Jagadis Bose National Talent Search (JBNSTS) Junior Scholarship*, 2022 funded by Department of Science and Technology and Biotechnology, Government of West Bengal.

Academic Interests

- Statistical Learning | Causal Inference | Probability Theory | Mathematical Statistics | Combinatorics

Talks and Presentations

- At ICTS, S N Bhatt Memorial Excellence Fellowship Programme: "Random Cluster Model on Random Regular Graphs".
- At Math Club ISI Kolkata: "A Probabilistic Proof of Cayley's Theorem using Poisson Branching Process".
- Presented a class project on "Different Clustering Methods and Comparison of their Performances in NCI60 Dataset".

Skills

- **Programming:** R, C, LaTeX, HTML.

Additional Information

- **Hobbies:** Singing, active trainee in Indian Raga Music, learning languages, watching movies, travelling, reading novels, learning instruments.
- **Extracurricular Activities:** Member of Core Committee in Math Club, ISI Kolkata, and Cultural Committee, ISI Kolkata.
- **Languages:** Bengali (native), Hindi, English (professional proficiency), and Italian (elementary proficiency).